Logistic regression

modelling binary decisions

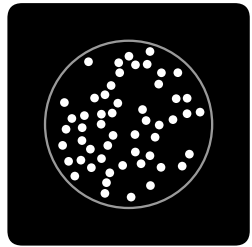
Alexandre Hyafil

Data Analysis for Cognitive Neuroscience





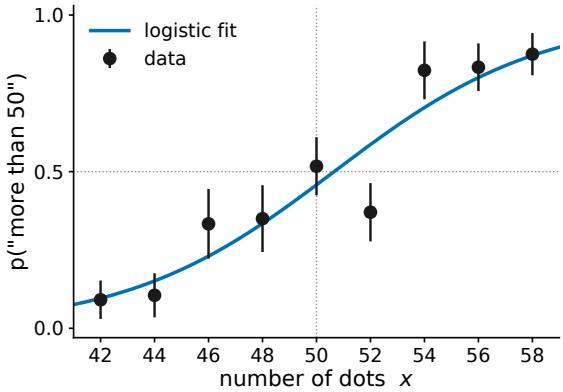
One stimulus: the psychometric curve

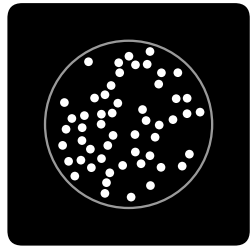


“more or less than 50 dots?”

$$p(y = 1 \mid x) = \sigma(w_0 + w_1 x)$$

One stimulus: the psychometric curve

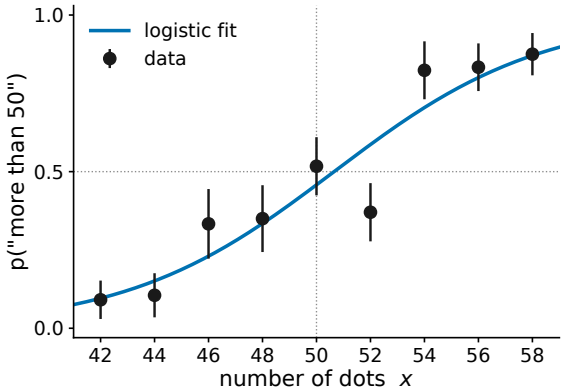


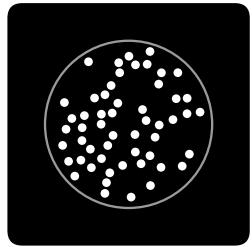


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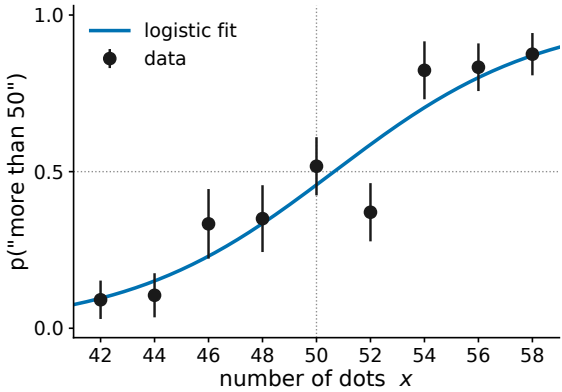


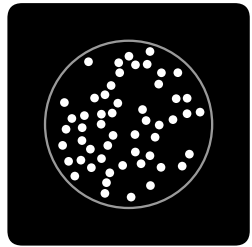
"more or less than 50 dots?"

$$p(y = 1 \mid x) = \sigma(w_0 + w_1 x)$$

- w_1 — the **slope**

One stimulus: the psychometric curve



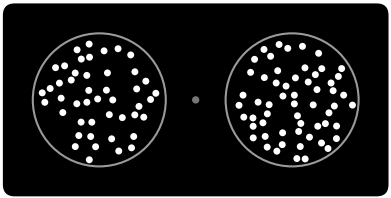


"more or less than 50 dots?"

$$p(y = 1 \mid x) = \sigma(w_0 + w_1 x)$$

- w_1 — the **slope**
- w_0 — the **bias**: indifference at $x = -w_0/w_1$

Two stimuli: a comparison task

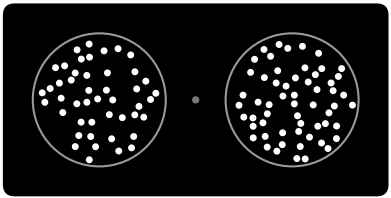


"which side has more dots?"

$x_L = 46$ $x_R = 58$

$$p(y = 1) = \sigma\big(w_0 + w_1\, x_L + w_2\, x_R\big)$$

Two stimuli: a comparison task



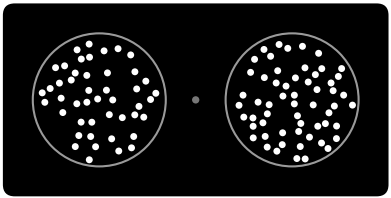
"which side has more dots?"

$x_L = 46$ $x_R = 58$

- $y = 1$ for *"more on the right"*

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Two stimuli: a comparison task



- $y = 1$ for *"more on the right"*

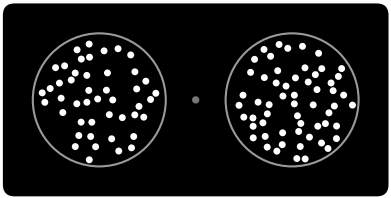
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bias

Two stimuli: a comparison task



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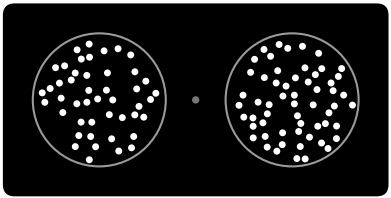
$x_L = 46$ $\quad$ $x_R = 58$

$$p(y = 1) = \sigma\big(w_0 + w_1\, x_L + w_2\, x_R\big)$$

bias

impact of the left stimulus

Two stimuli: a comparison task



- $y = 1$ for *"more on the right"*

"which side has more dots?"

$x_L = 46$ $x_R = 58$

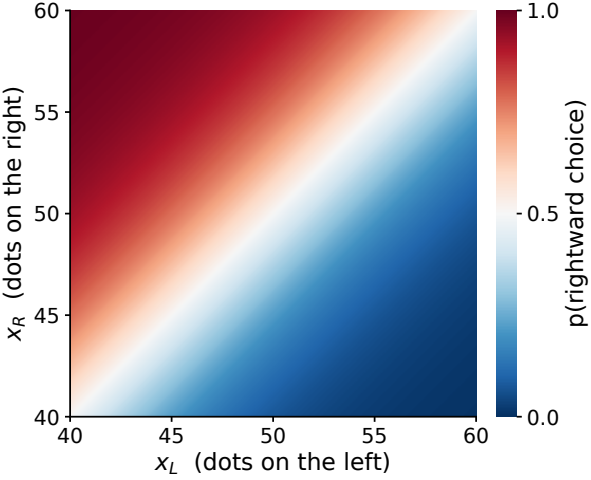
$$p(y = 1) = \sigma\big(w_0 + w_1 \, x_L + w_2 \, x_R \big)$$

bias

impact of the left stimulus

impact of the right stimulus

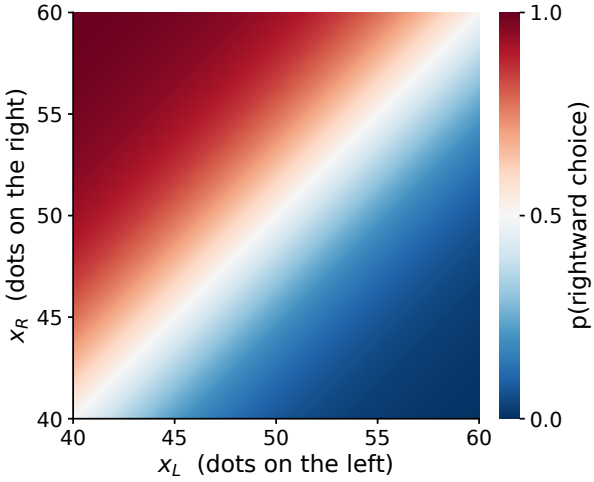
Two stimuli: a comparison task



$$\sigma\big(w_0 + w_1 x_L + w_2 x_R\big)$$

- $w_1 = -0.25, \quad w_2 = +0.25,$ $w_0 = 0$

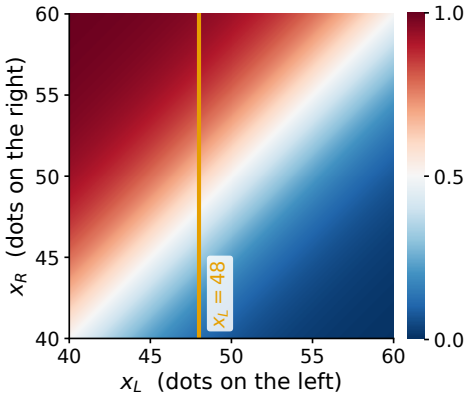
Two stimuli: a comparison task

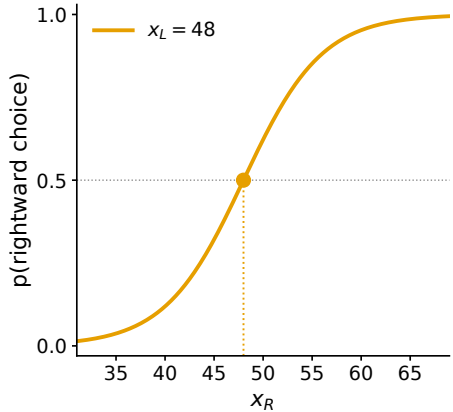


$$\sigma\big(w_0 + w_1 x_L + w_2 x_R\big)$$

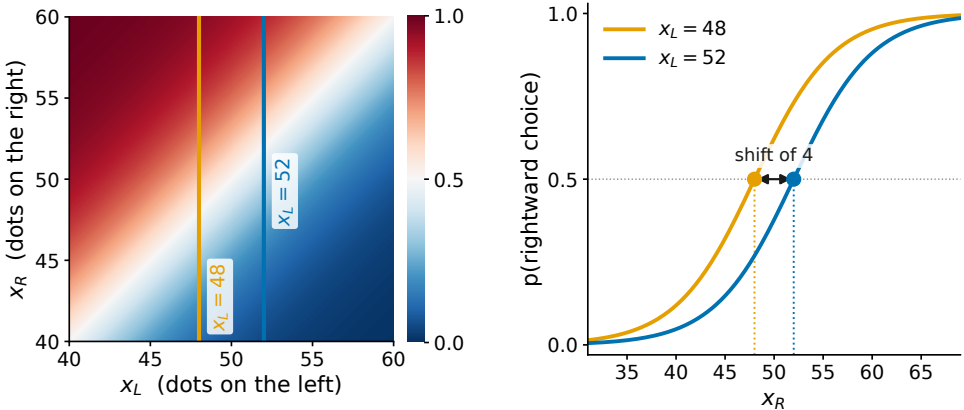
- $w_1 = -0.25, \quad w_2 = +0.25, \quad w_0 = 0$
- the white band is where the subject is at chance

Fix x_L: a psychometric curve for x_R



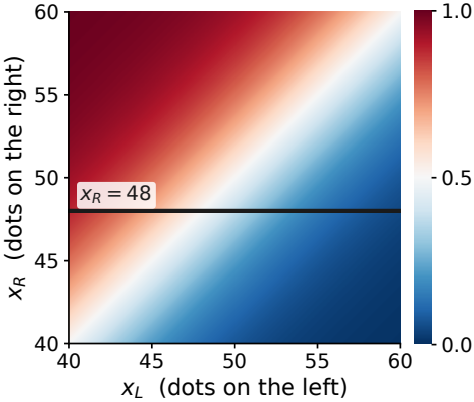


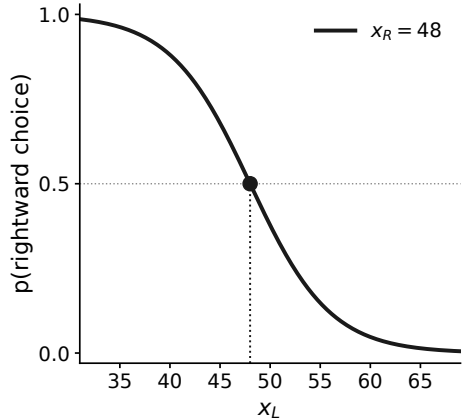
Fix x_L: a psychometric curve for x_R



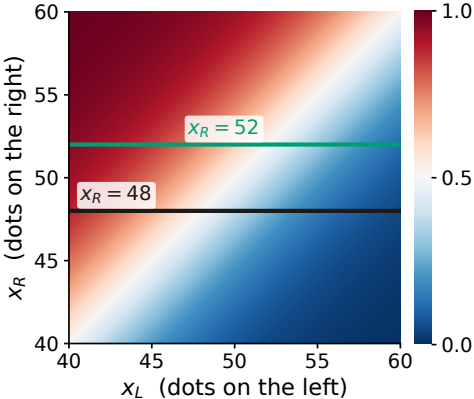
- same slope, shifted: the curve always crosses 0.5 at $x_R = x_L$

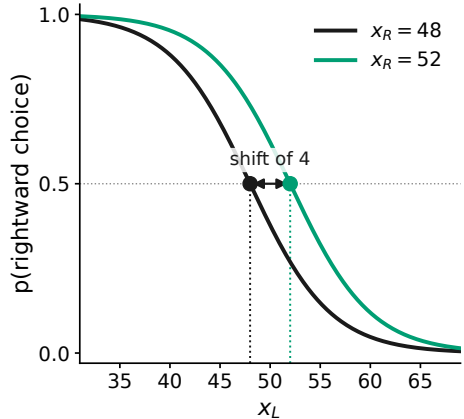
Fix x_R: the curve runs the other way



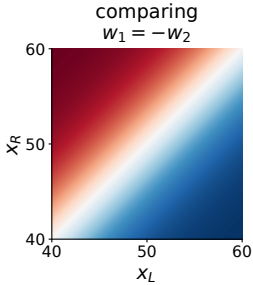


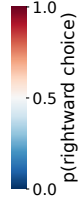
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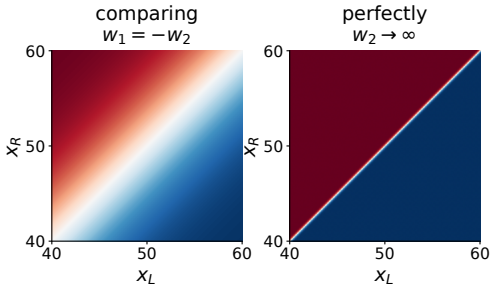


The weights are the subject's strategy

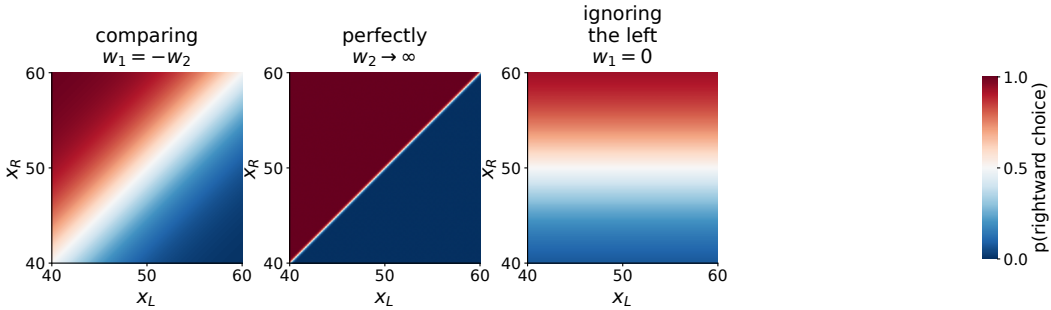




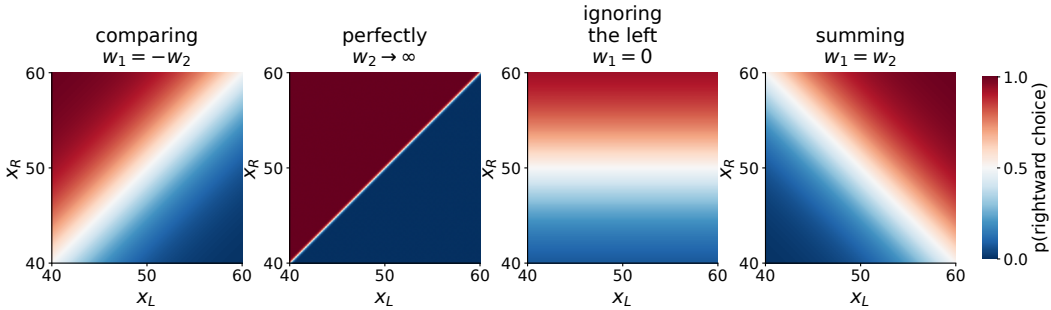
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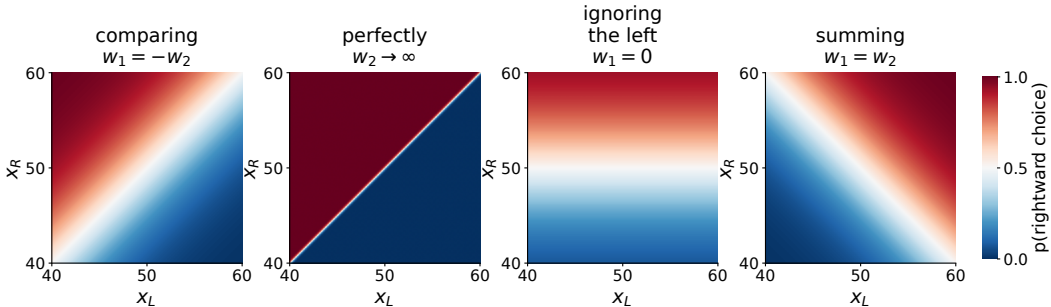
The weights are the subject's strategy



The weights are the subject's strategy

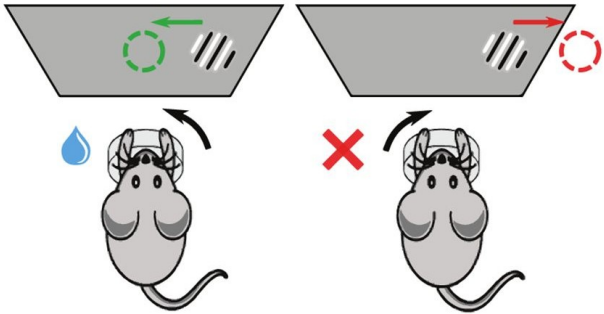


The weights are the subject's strategy



The *orientation* of the bands gives the strategy away.

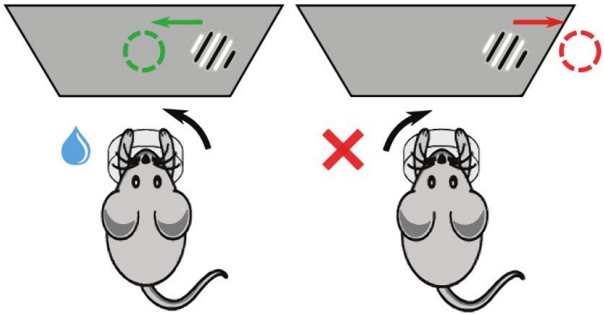
A real example: the IBL mouse task



- a grating on the left or the right, at one of several contrasts

International Brain Laboratory et al., *eLife* 10:e63711 (2021), CC BY 4.0

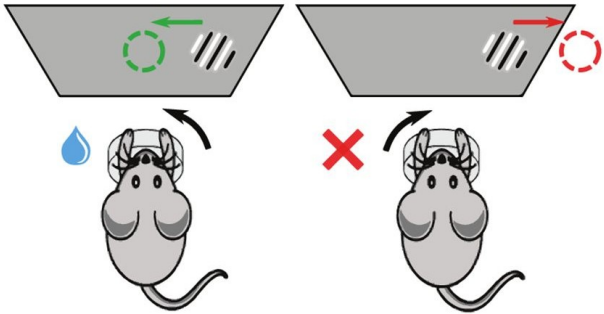
A real example: the IBL mouse task



- a grating on the left or the right, at one of several contrasts
- the mouse turns a wheel to bring it to the centre

International Brain Laboratory et al., *eLife* 10:e63711 (2021), CC BY 4.0

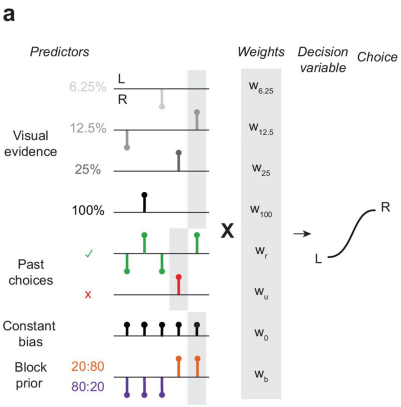
A real example: the IBL mouse task



- a grating on the left or the right, at one of several contrasts
- the mouse turns a wheel to bring it to the centre
- correct $\Rightarrow$ water

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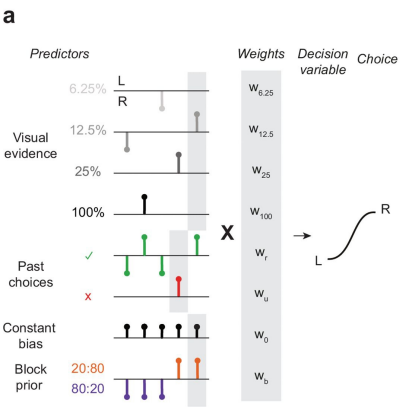
Logistic regression on the mouse's choices

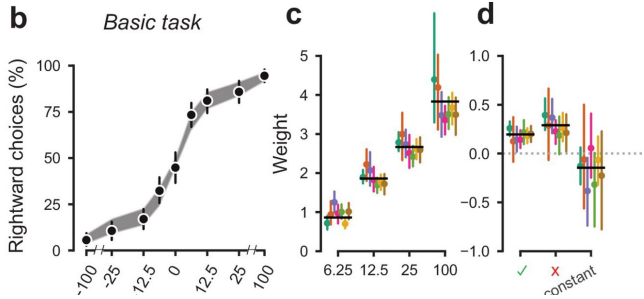


- predictors: visual evidence, **previous choices**, constant bias

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Logistic regression on the mouse's choices

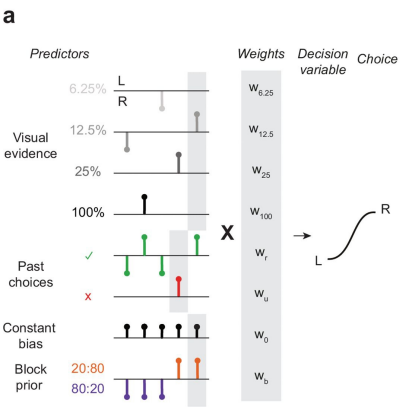


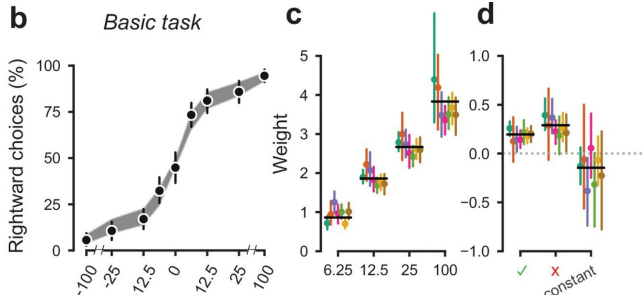


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Logistic regression on the mouse's choices

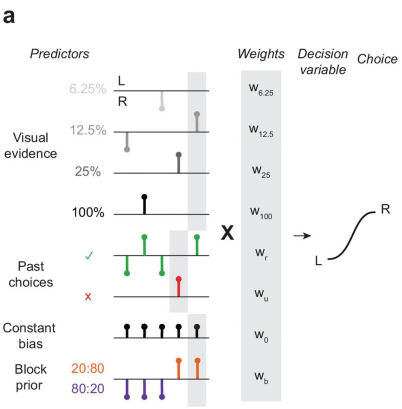


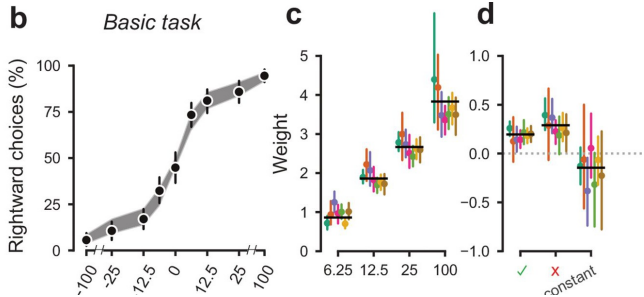


- predictors: visual evidence, **previous choices**, constant bias
- the weight on visual evidence grows with contrast

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Logistic regression on the mouse's choices





- predictors: visual evidence, **previous choices**, constant bias
- the weight on visual evidence grows with contrast
- past choices carry *positive* weights whether rewarded or not — a **win-stay–lose-stay** strategy

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Two directions: encoding and decoding



Encoding

predict *neural activity* from the cognitive variables

Two directions: encoding and decoding



Encoding

predict *neural activity* from the cognitive variables

Decoding

predict the *cognitive variables* from the neural activity

Two directions: encoding and decoding



Encoding

predict *neural activity* from the cognitive variables

+ closer to the causal mechanism

Decoding

predict the *cognitive variables* from the neural activity

– can succeed on information the brain never uses

Two directions: encoding and decoding



Encoding

predict *neural activity* from the cognitive variables

- \+ closer to the causal mechanism
- \+ confounds can be entered as regressors and factored out

Decoding

predict the *cognitive variables* from the neural activity

- – can succeed on information the brain never uses
- – confounds are harder to rule out

Two directions: encoding and decoding



Encoding

predict *neural activity* from the cognitive variables

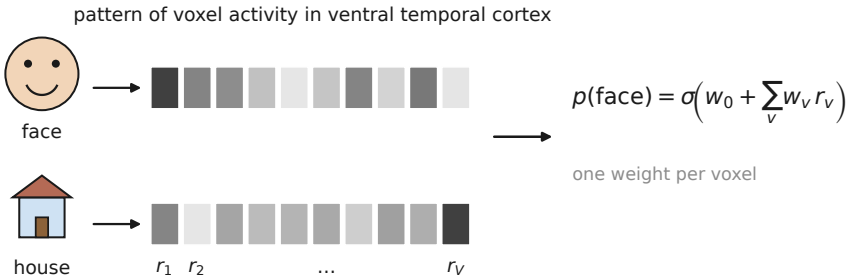
- \+ closer to the causal mechanism
- \+ confounds can be entered as regressors and factored out
- – one voxel / neuron at a time: ignores the population

Decoding

predict the *cognitive variables* from the neural activity

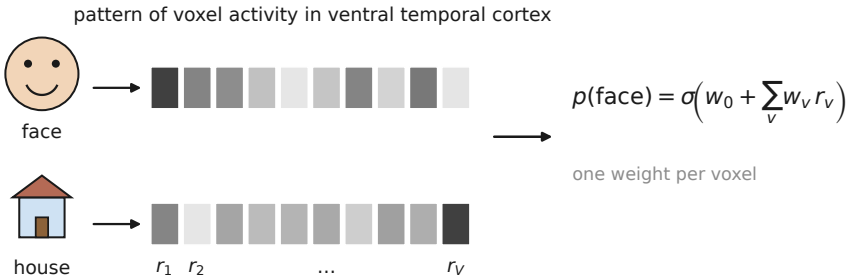
- – can succeed on information the brain never uses
- – confounds are harder to rule out
- \+ reads out the whole population at once

Decoding with logistic regression



Face/house paradigm: Haxby et al., *Science* 293:2425 (2001) — who classified by pattern correlation, not logistic regression.

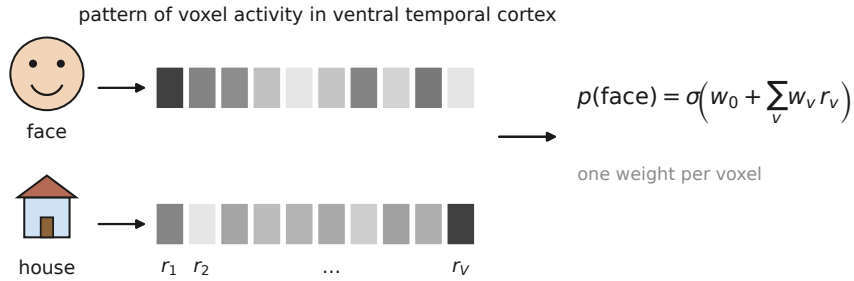
Decoding with logistic regression



- same equation — but the predictors r_v are now **neural**, not experimental

Face/house paradigm: Haxby et al., *Science* 293:2425 (2001) — who classified by pattern correlation, not logistic regression.

Decoding with logistic regression



- same equation — but the predictors r_v are now **neural**, not experimental
- above-chance accuracy on held-out trials ⇒ the category is represented there

Face/house paradigm: Haxby et al., *Science* 293:2425 (2001) — who classified by pattern correlation, not logistic regression.